Section 7

AND

0 and 0 = 0

0 and 1 = 0

1 and 0 = 0

1 and 1 = 1

OR

0 or 0 = 0

0 or 1 = 1

1 or 0 = 1

1 or 1 = 1

XOR

0 xor 0 = 0

0 xor 1 = 1

1 xor 0 = 1

1 xor 1 = 0

Question 1

For the following instructions; indicate the value of the result register, and also give the AF, CF, and ZF

Q1

AX = F000, BX = 3456, and DX = E390

A) AND DX, AX <-> DX=DX and AX

1110 0011 1001 0000 CF=0, ZF=0

1111 0000 0000 0000 AF unaffected

1110 0000 0000 0000

DX = E000

Q1

AX = F000, BX = 3456, and DX = E390

B) OR DH, BL

1110 0011

0101 0110 CF=0, ZF=0

1111 0111

DH = F7

Q1

AX = F000, BX = 3456, and DX = E390

C)XOR AL, 76H

0000 0000

0111 0110 CF=0,ZF=0

0111 0110

AL = 76H

Q1

AX = F000, BX = 3456, and DX = E390

D) AND DX, DX

1110 0011 1001 0000

1110 0011 1001 0000 CF=0,ZF=0

1110 0011 1001 0000

DX=E390

Q1

AX = F000, BX = 3456, and DX = E390

E) XOR AX, AX

1111 0000 0000 0000

1111 0000 0000 0000 CF=0,ZF=1

0000 0000 0000 0000

AX = 0

Q1

AX = F000, BX = 3456, and DX = E390

G) AND AH, 0FF

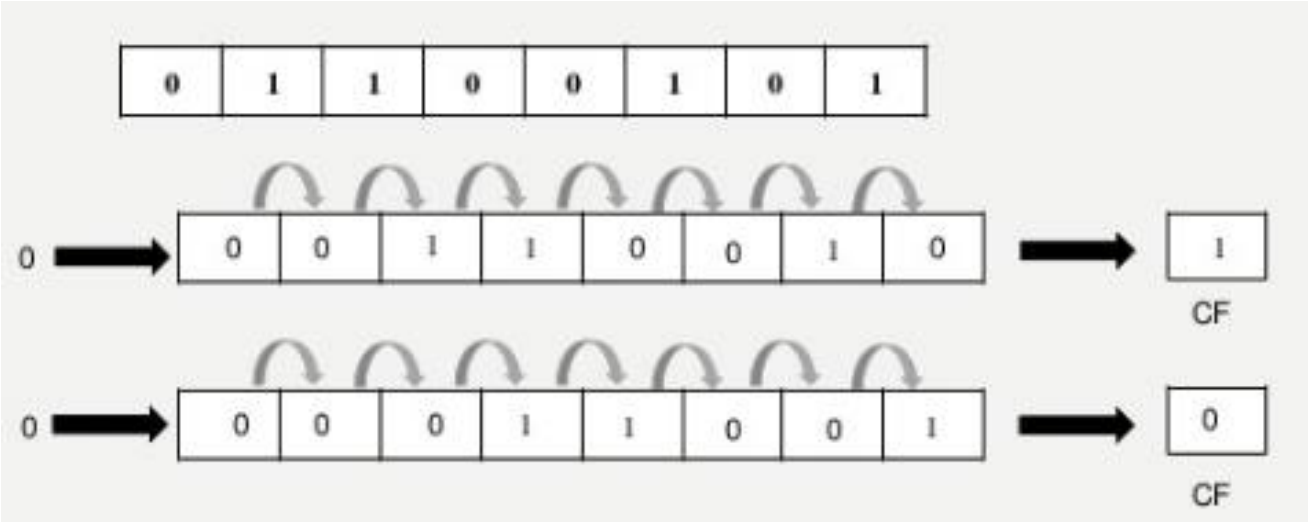
1111 0000

1111 1111 CF=0,ZF=0

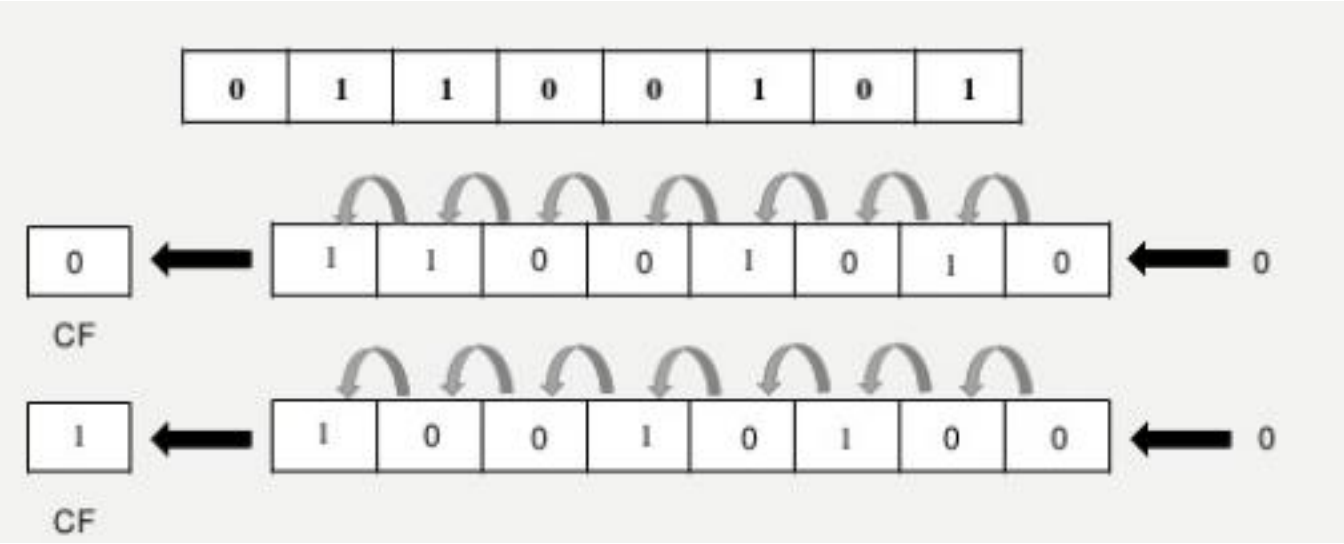
1111 0000

AH=F0

Shift Right SHR



Shift Left SHL



Q1

AX = F000, BX = 3456, and DX = E390

K)MOV CL, 04

SHL AL, CL

CF=0 0000 0000 0 0 0 0

ZF=1

AL=0

Q1

AX = F000, BX = 3456, and DX = E390

L)SHR DX, 1

0 → 1110001110010000 → CF

0111000111001000 CF=0,ZF=0

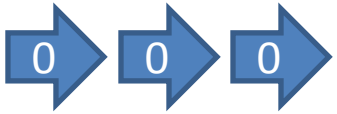
DX=71C8

Q1

AX = F000, BX = 3456, and DX = E390

M)MOV CL, 3

SHR DL, CL



10010000

00010010 CF=0, ZF=0

DL=12H

ROR (rotate right)

In rotate right, as bits are shifted from left to right they exit from the right end (LSB) and enter the left end (MSB). In addition, as each bit exits the LSB, a copy of it is given to the carry flag. In other words, in ROR the LSB is moved to the MSB and is also copied to CF, as shown in the diagram. If the operand is to be rotated once, the 1 is coded, but if it is to be rotated more than once, register CL is used to hold the number of times it is to be rotated.

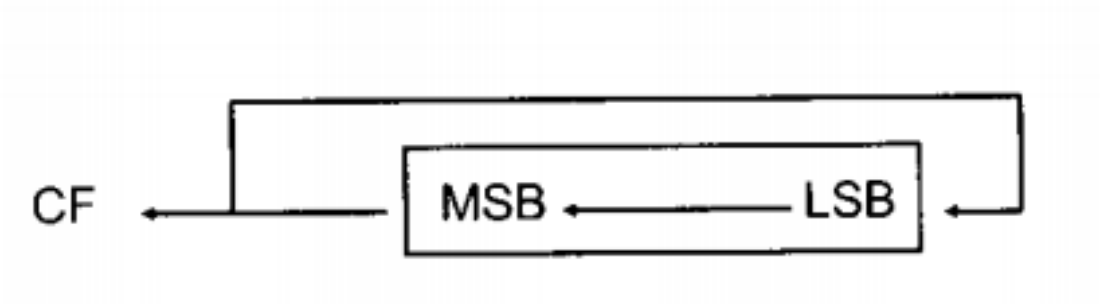
Example ROR

```
MOV   AL,36H      ;AL=0011 0110
ROR   AL,1        ;AL=0001 1011  CF=0
ROR   AL,1        ;AL=1000 1101  CF=1
ROR   AL,1        ;AL=1100 0110  CF=1
```

or:

```
MOV   AL,36H      ;AL=0011 0110
MOV   CL,3        ;CL=3 number of times to rotate
ROR   AL,CL       ;AL=1100 0110 CF=1
```

ROL (rotate left)



In rotate left, as bits are shifted from right to left they exit the left end (MSB) and enter the right end (LSB). In addition, every bit that leaves the MSB is copied to the carry flag. In other words, in ROL the MSB is moved to the LSB and is also copied to CF, as shown in the diagram. If the operand is to be rotated once, the 1 is coded. Otherwise, the number of times it is to be rotated is in CL.

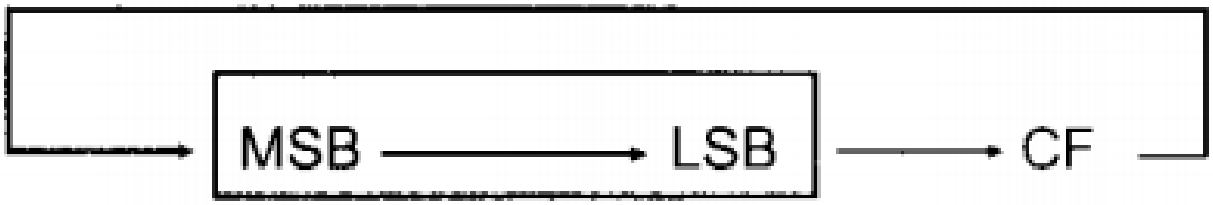
Example ROL

```
MOV     BH,72H          ;BH=0111 0010
ROL     BH,1            ;BH=1110 0100  CF=0
ROL     BH,1            ;BH=1100 1001  CF=1
ROL     BH,1            ;BH=1001 0011  CF=1
ROL     BH,1            ;BH=0010 0111  CF=1
```

The operand can be a word:

```
MOV     DX,672AH        ;DX=0110 0111 0010 1010
MOV     CL,3            ;CL=3 number of times to rotate
ROL     DX,CL           ;DX=0011 1001 0101 0011 CF=1
```

RCR (rotate carry right)



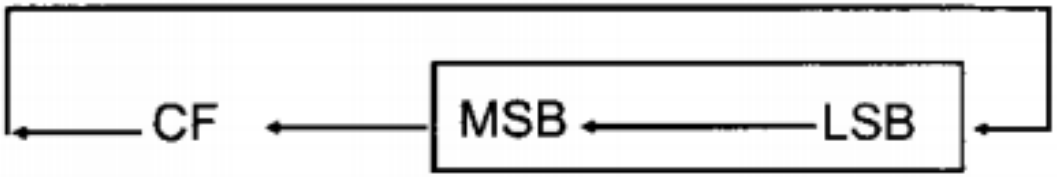
In RCR, as bits are shifted from left to right, they exit the right end (LSB) to the carry flag, and the carry flag enters the left end (MSB). In other words, in RCR the LSB is moved to CF and CF is moved to the MSB. In reality, CF acts as if it is part of the operand. This is shown in the diagram. If the operand is to be rotated once, the 1 is coded, but if it is to be rotated more than once, the register CL holds the number of times.

RCR example

```
CLC                     ;make CF=0
MOV     AL,26H          ;AL=0010 0110
RCR     AL,1            ;AL=0001 0011 CF=0
RCR     AL,1            ;AL=0000 1001 CF=1
RCR     AL,1            ;AL=1000 0100 CF=1
```

RCL (rotate carry left)

In RCL, as bits are shifted from right to left they exit the left end (MSB) and enter the carry flag, and the carry flag enters the right end (LSB). In other words, in RCL the MSB is moved to CF and CF is moved to the LSB. In reality, CF acts as if it is part of the operand. This is shown in the diagram. If the operand is to be rotated once, the 1 is coded, but if it is to be rotated more than once, register CL holds the number of times.



RCL example

```
STC                     ;make CF=1
MOV     BL,15H          ;BL=0001 0101
RCL     BL,1            ;0010 1011 CF=0
RCL     BL,1            ;0101 0110 CF=0
```

CMP destination, source

Table 3-3: Flag Settings for Compare Instruction

Compare operands	CF	ZF
destination > source	0	0
destination = source	0	1
destination < source	1	0

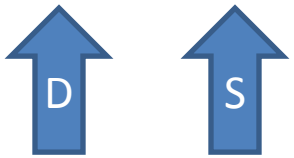
Question 2

Give the values of CF and ZF after the following:

Q2

a) MOV BX, 2500

CMP BX, 1400



Greater CF=0, ZF=0

Q2

b) MOV DL, 34
CMP DL, 88

Less CF=1, ZF=0

Q2

c) SUB AX, AX	Ax=0
CMP AX, 0000	Equal CF=0, ZF=1
CMP AX, 0FFFFH	Less CF=1, ZF=0

CMP destination, source

Table 3-3: Flag Settings for Compare Instruction

Compare operands	CF	ZF
destination > source	0	0
destination = source	0	1
destination < source	1	0

Mnemonic	Condition Tested	"Jump IF ..."
JA/JNBE	(CF = 0) and (ZF = 0)	above/not below nor zero
JAE/JNB	CF = 0	above or equal/not below
JB/JNAE	CF = 1	below/not above nor equal
JBE/JNA	(CF or ZF) = 1	below or equal/not above
JC	CF = 1	carry
JE/JZ	ZF = 1	equal/zero
JG/JNLE	((SF xor OF) or ZF) = 0	greater/not less nor equal
JGE/JNL	(SF xor OF) = 0	greater or equal/not less
JL/JNGE	(SF xor OR) = 1	less/not greater nor equal
JLE/JNG	((SF xor OF) or ZF) = 1	less or equal/not greater
JNC	CF = 0	not carry
JNE/JNZ	ZF = 0	not equal/not zero
JNO	OF = 0	not overflow
JNP/JPO	PF = 0	not parity/parity odd
JNS	SF = 0	not sign
JO	OF = 1	overflow
JP/JPE	PF = 1	parity/parity equal
JS	SF = 1	sign

Question 3

Indicate if the jump will happen or not:

A)

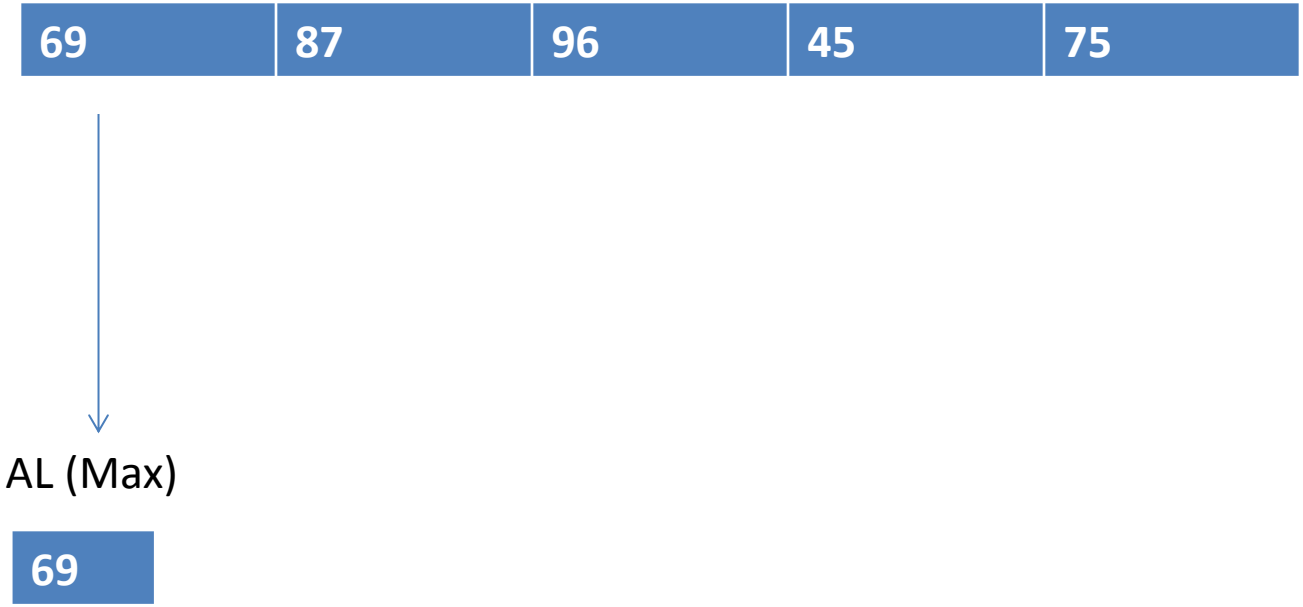
```
MOV CL, 5
SUB AL, AL          ;AL=00000000
SHL AL, CL          ;CF=0
JNC TARGET          ; jump happens
```

Q3

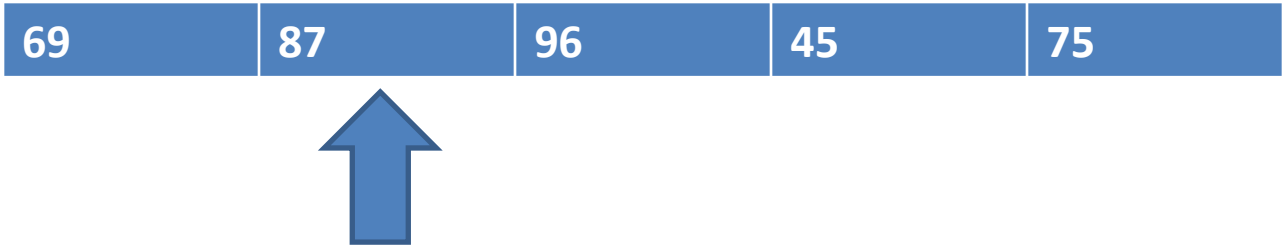
```
MOV BX, 2500
CMP BX, 1400
JA TARGET                    ; jump happens
```

Write an assembly program that finds the maximum number in an array

Maximum number in an array



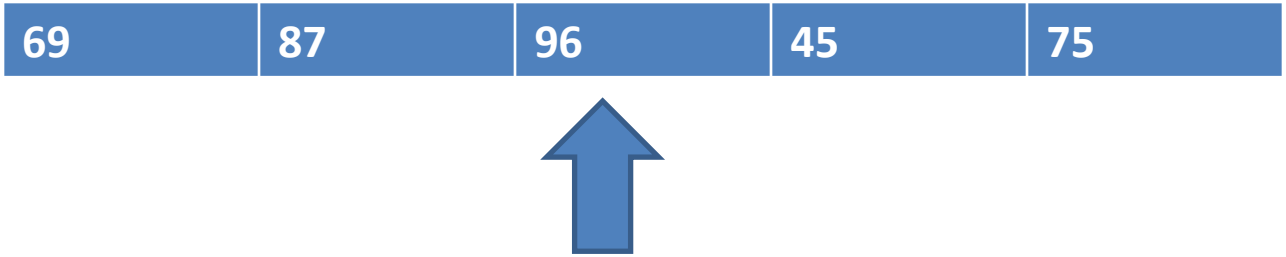
Maximum number in an array



If Greater than Max
Will be the new Max

AL (Max)

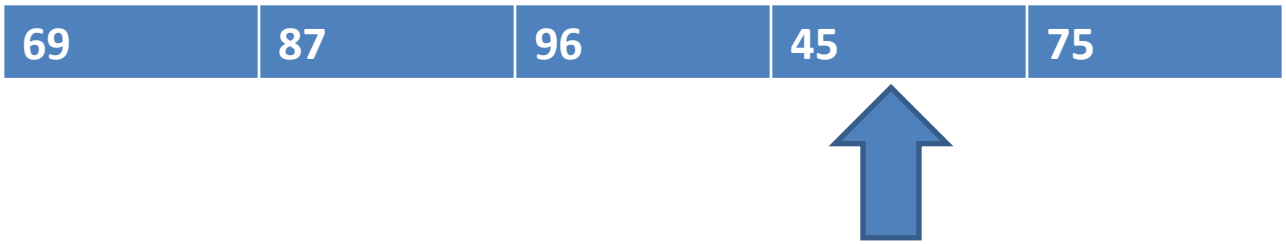
Maximum number in an array



If Greater than Max
Will be the new Max

AL (Max)

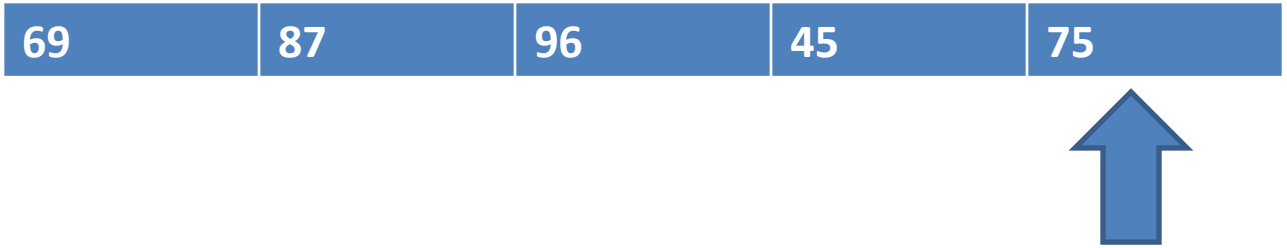
Maximum number in an array



If Greater than Max
Will be the new Max

AL (Max)

Maximum number in an array



If Greater than Max
Will be the new Max

AL (Max)

Write an assembly program that turns Uppercase letters into lowercase

ASCII Table

Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value
00	NUL	10	DLE	20	SP	30	0	40	@	50	P	60	`	70	p
01	SOH	11	DC1	21	!	31	1	41	A	51	Q	61	a	71	q
02	STX	12	DC2	22	"	32	2	42	B	52	R	62	b	72	r
03	ETX	13	DC3	23	#	33	3	43	C	53	S	63	c	73	s
04	EOT	14	DC4	24	$	34	4	44	D	54	T	64	d	74	t
05	ENQ	15	NAK	25	%	35	5	45	E	55	U	65	e	75	u
06	ACK	16	SYN	26	&	36	6	46	F	56	V	66	f	76	v
07	BEL	17	ETB	27	'	37	7	47	G	57	W	67	g	77	w
08	BS	18	CAN	28	(	38	8	48	H	58	X	68	h	78	x
09	HT	19	EM	29	)	39	9	49	I	59	Y	69	i	79	y
0A	LF	1A	SUB	2A	*	3A	:	4A	J	5A	Z	6A	j	7A	z
0B	VT	1B	ESC	2B	+	3B	;	4B	K	5B	[	6B	k	7B	{
0C	FF	1C	FS	2C	,	3C	<	4C	L	5C	\	6C	l	7C	|
0D	CR	1D	GS	2D	-	3D	=	4D	M	5D	]	6D	m	7D	}
0E	SO	1E	RS	2E	.	3E	>	4E	N	5E	^	6E	n	7E	~
0F	SI	1F	US	2F	/	3F	?	4F	O	5F	_	6F	o	7F	DEL

Write an assembly program that counts the number of zero in a binary word

Numbers of zero in a word

0111000011110000

CF

Numbers of zero in a word

0011100001111000

CF=0

Numbers of zero in a word

0001110000111100

CF=0

Numbers of zero in a word

0000111000011110

CF=0

Numbers of zero in a word

0000011100001111

CF=0

Numbers of zero in a word

0000001110000111

CF=1

Numbers of zero in a word

0000000111000011

CF=1

Numbers of zero in a word

0000000011100001

CF=1

Numbers of zero in a word

0000000001110000

CF=1

Numbers of zero in a word

0000000000111000

CF=0

Numbers of zero in a word

0000000000011100

CF=0

Numbers of zero in a word

0000000000001110

CF=0

Numbers of zero in a word

0000000000000111

CF=0

Numbers of zero in a word

0000000000000011

CF=1

Numbers of zero in a word

0000000000000001

CF=1

Numbers of zero in a word

0000000000000000

CF=1